Absorbance and Concentration · One complete topic

Formulas and symbols

Every formula in scope, then a glossary of every symbol used.

Exactly the definitions, procedure, numerical or code example and limitations in the included lesson · compiled 17 October 2026

Contents

1. Part A · Formula sheet
2. 1 · Absorbance and Concentration
3. Part B · Symbol glossary

A Formula sheet

1 Absorbance and Concentration

E1.1 · Beer–Lambert relation

$$A = \varepsilon\, l\, c = -\log_{10} T$$

gives	Absorbance of a dilute absorbing species, such as illustrative $\mathrm{Cu^{2+}}$ in solution.
applies	Monochromatic measurement in the linear range with a suitable blank and consistent units.
units	A: dimensionless; ε: $\mathrm{L\,mol^{-1}\,cm^{-1}}$; l: cm; c: $\mathrm{mol\,L^{-1}}$.

S1

B Symbol glossary

Symbol	Name	Meaning	Unit	Appears in
A	Absorbance	Negative base-ten logarithm of fractional transmittance. S1	Dimensionless	E1.1
ε	Molar absorption coefficient	Proportionality coefficient at the specified species, wavelength and conditions. S1	$\mathrm{L\,mol^{-1}\,cm^{-1}}$	E1.1

Symbol	Name	Meaning	Unit	Appears in
l	Optical path length	Distance light travels through the sample. S1	cm	E1.1
c	Concentration	Amount of absorbing species per unit solution volume. S1	$\mathrm{mol\,L^{-1}}$	E1.1
T	Fractional transmittance	Ratio I/I_0; use the fraction, not its percentage. S1	Dimensionless	E1.1, W1.1
I, I_0	Transmitted and incident intensities	Both intensities use the same wavelength, geometry and units; their ratio is dimensionless. S1	Same intensity unit	E1.1